

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech (CL) Examination****March-April 2018****CL434 Urban Transportation System****Date: 06.04.2018, Friday****Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1** From an in-out survey conducted for a parking area consisting of 50 bays, the initial count was found to be 35. The number of vehicles coming in and out of the parking lot for a time interval of 5 minutes is as shown in the Table 1. Find the Parking accumulation, Total parking load, Average occupancy and Efficiency of the parking lot. **[07]**

Table 1: In-Out Survey

Time (min)	In	Out	Time (min)	In	Out
5	3	2	35	4	3
10	2	4	40	2	4
15	4	3	45	1	2
20	7	4	50	5	4
25	6	5	55	3	1
30	5	2	60	6	3

- Q – 2.a** What are the measures to be taken for traffic improvement or management? **[04]**
- Q – 2.b** What are the functions or role of Transportations at national, regional and urban level? **[03]**
- Q – 2.c** What do you mean by Urbanization? What are the push and pull factors affecting the growth of urbanization? Discuss in detail about the various types of parking methods with neat sketch. **[07]**

OR

- Q – 2.c** Explain with the help of flow diagram about the Urbanization cycle. Discuss in detail about the various survey technique used for doing the parking study. **[07]**
- Q - 3** **Answer any one.** **[14]**
- a.** Write down the various methods used for trip distribution model and describe in detail any two methods. A study area is divided into four zones 1, 2, 3 & 4. The present trip

distribution and future trips are shown in Table 2. Develop a future trip distribution matrix using Detroit method.

Table 2: Trip Distribution

O \ D	1	2	3	4	Total Present trips	Future Trips
1	-	50	60	30	140	280
2	40	-	70	20	130	390
3	20	60	-	40	120	300
4	50	70	30	-	150	220
Total Present trips	110	180	160	90	-	-
Future Trips	200	500	340	150	-	-

- b. The probability of choosing the car mode (P_c) is found to be given :

$$P_c = (1/1+e^{-u_x}) \text{ where , } u_x = 0.70-0.04(tt_{car}-tt_{bus})$$

The total trip exchange between zones are given in Table 3 and travel time value is given in Table 4

Table 3 Zonal trip exchange

From	To	Person trips/day
A	B	1200
B	A	0
A	C	800
C	A	1600
B	C	900
C	B	800

Table 4 Travel time

Zones	Travel time in minutes	
	Car	Bus
A-B	15	20
B-C	8	12

Determine two way volume in cars per day on the road AB and BC and if the average occupancy of car is 2.6.

SECTION – II**Q - 4 Answer the question below.****[07]**

- (1) At what distance apart from intersection, Busbays should be located?
 a) 55m b) 90m c) 75m d) 80m
- (2) Maximum design speed for Arterial road according to IRC 86.
 a) 55 kmph b) 65 kmph c) 80 kmph d) 100 kmph
- (3) Recommended carriageway width for 2-lane road without curb
 a) 6.5m b) 7.0m c) 7.5m d) 9m
- (4) What is diverging traffic in rotary?
- (5) Write down the equation for width of weaving section.
- (6) Rotary is suitable when major proportion of traffic is:
 a) Straight b) Right turning c) Left turning d) U-turn
- (7) The minimum width for the cycle track is:
 a) 2 m b) 2.5 m c) 3 m d) 4.5m

Q – 5.a Define: (1) Saturation flow (2) Cycle length (3) Load Factor.**[03]****OR****Q – 5.a Explain rotary design element: (1) Entry exit radius (2) Radius of central island****[03]****Q – 5.b Discuss the standard of design speed and footpath (side walk) according to IRC guideline for urban road.****[04]****Q – 5.c Sketch the basic forms of At-grade intersections and also explain basic principles of design for At-grade intersection.****[07]****OR****Q – 5.c Explain the warrants for deciding the Signal installation at Intersection.****[07]****Q – 6.a The width of approaches for a rotary intersection is 12 m. The entry and exit width at the rotary is 10 m. Table below gives the traffic from the four approaches, traversing the intersection. Find the capacity of the rotary.****[10]**

Approach	Left turn	Straight	Right turn
North	400	700	300
South	350	370	420
East	200	450	550
West	350	500	520

Q – 6.b Write down objective and requirement of Transportation System Management planning. **[04]**

OR

Q – 6.a Write down different TSM actions and Explain about the Ride share and Road Pricing. **[05]**

Q – 6.b What are the benefits of TSM system for traffic improvement strategies? **[05]**

Q – 6.c Explain by sketch Merging, Diverging and Weaving flow. **[04]**
